ACM CCS'26 Artifact Appendix: From Snapshot to Snooping: An Empirical Study on Geolocation Privacy Leakage in Large Vision Language Models

Yihe Zhou
City University of Hong Kong
Hong Kong
yihezhou2@cityu.edu.hk

Tao Ni
King Abdullah University of Science and Technology
Saudi Arabia
tao.ni@kaust.edu.sa

Qingchuan Zhao*
City University of Hong Kong
Hong Kong
qizhao@cityu.edu.hk

Cong Wang
City University of Hong Kong
Hong Kong
congwang@cityu.edu.hk

A Artifact Appendix

A.1 Abstract

This Artifact Appendix presents the code artifact of image crawling, selection and filtering, model querying, geolocation performance evaluation, and ablation studies in our empirical study on geolocation privacy leakage in Large Vision Language Models. This artifact enables reproduction of the main experiments in our evaluation pipeline.

A.2 Description & Requirements

A.2.1 Security, privacy, and ethical concerns. The artifact does not pose risks to the evaluator's machine security or data privacy, nor does it raise ethical concerns. In addition, the image resources we use for evaluation are all publicly available benchmark datasets or accessible from public resources. We do not disclose per-record geolocation inference results for each image, only the aggregate final results in the main text. We also do not disclose the complete Flickr6K dataset crawled from Flickr for privacy concerns; sample images from Flickr can be found in `/dataset/Flickr6k`, and Flickr ground-truth geolocation can be found in `flickr6k_geocode.csv`. YFCC4K ground truth can be found in `yfcc4k_gt_geocode.csv`.

A.2.2 How to access. The artifact is available at https://doi.org/10.5281/zenodo.23764475. Please download the latest version.

A.2.3 Hardware dependencies. None.

A.2.4 Software dependencies. This artifact is implemented in Python (Python 3.9.24 recommended). All required dependencies are specified in `requirements.txt`.

A.2.5 Benchmarks. Our work uses the public geolocation benchmark dataset YFCC4K, available at http://www.mediafire.com/file/3og8y3o6c9de3ye/yfcc4k.zip. The additional Flickr6k dataset can be crawled using /code/retriever.py. The ground-truth geolocation of our datasets is available in the `dataset` folder.

A.3 Set Up

A.3.1 Installation. This artifact is implemented in Python (Python 3.9.24 recommended). All required dependencies are specified in `requirements.txt`.

A.3.2 Basic test. Run `query_shot.py`, set OpenRouter API key and fill in the corresponding model code from OpenRouter, set `IMAGE_FOLDER`, `OUTPUT_CSV` and `PROPMT_TEXT` (in prompt.md), including the formatting instructions, and query the models.

For a quick evaluation of our existing predictions, please refer to `sampled_responses_with_categories.csv`, which includes three files of sampled responses from three of the LVLMs under both zero-shot and CoT prompting. They can be used to promptly reproduce Table 2 in the main paper.

The model codes we used for evaluation are as follows; note that some model might be unavailable by the time this artifact is published. Availability and model input and output prices per million tokens are detailed in README.md. They can be changed over time; please refer to real-time information on OpenRouter.

- `anthropic/claude-sonnet-4.5`
- `anthropic/claude-opus-4.6`
- `anthropic/claude-opus-4.8`
- `google/gemini-2.5-flash-image`
- `google/gemini-3.1-pro-preview`
- `mistralai/mistral-medium-3.1`
- `mistralai/mistral-small-2603`
- `x-ai/grok-4`
- `x-ai/grok-4.20`
- `x-ai/grok-4.6`
- `meta-llama/llama-4-scout`
- `meta-llama/llama-4-maverick`
- `qwen/qwen2.5-vl-72b-instruct`
- `qwen/qwen3.5-397b-a17b`
- `openai/gpt-4o-mini`
- `openai/o3`
- `openai/gpt-5.5`
- `minimax/minimax-m3`

*The corresponding author.

- `moonshotai/kimi-k2.7-code`

A.3.3 Filtering and Selection. Run `filter.py` to filter out low-resolution images or those with no geolocation clues that the model rejects.

A.4 Evaluation Workflow

The models should be prompted with the exact format instructions as: 1. LATITUDE: xxx; 2. LONGITUDE: xxx; 3. LOCATION: City, Country, and Continent; (4. REASONING: xxx). All prediction files should strictly follow the column format: filename, latitude, longitude, city, country, continent, (and reasoning) before evaluating with ground truth. In practice, some model outputs may not follow the given format; manual handling will be required to convert raw responses to the above-mentioned format. If all raw predictions follow the given format, `code/conversion-0shot` and `conversion_cot` can be used to format the columns.

A.4.1 Distance-based Evaluation. Using `eval/eval.py`, fill in the prediction (`predicted_file`), ground truth CSV path (`ground_truth_file`), and obtain the corresponding distance-threshold accuracy results, mean and maximum GeoScore, mean and minimum Haversine Distance.

A.4.2 Matching-based Evaluation. Use `eval.py` or `eval-match.py`, fill in the prediction (`PRED_CSV`), ground truth CSV file path (`GT_CSV`) and output CSV file path (`OUT_CSV`), and obtain the distance-based evaluation results or the matching-based evaluation results at the three levels: city, country, and continent.

A.4.3 Per-category Analysis, Continental, and Country-level. We used LLM labeling and human annotation to obtain the final 13-dimensional category labels, provided in `sampled_responses_with_categories.csv`. To obtain the evaluation results for images from a single category, use grouped labels in Excel (outdoor/indoor, continent, country, etc.) to filter out other predictions and run `eval.py`.

A.4.4 Safety Filter Test. Run `safety_test.py` with the five levels of prompts that are already included in the script. The labeling rule is detailed in README.md.

A.4.5 Ablation Studies. Run `res_degrade.py` or `grayscale_conversion.py` to degrade the original images to downscaled resolutions, or from RGB to grayscale. For testing the effect of prompting language, simply translate the English version `PROMPT_TEXT` to other languages using Google Translate, and query the model.

A.5 Version

Based on the LaTeX template for Artifact Evaluation V20220926.